



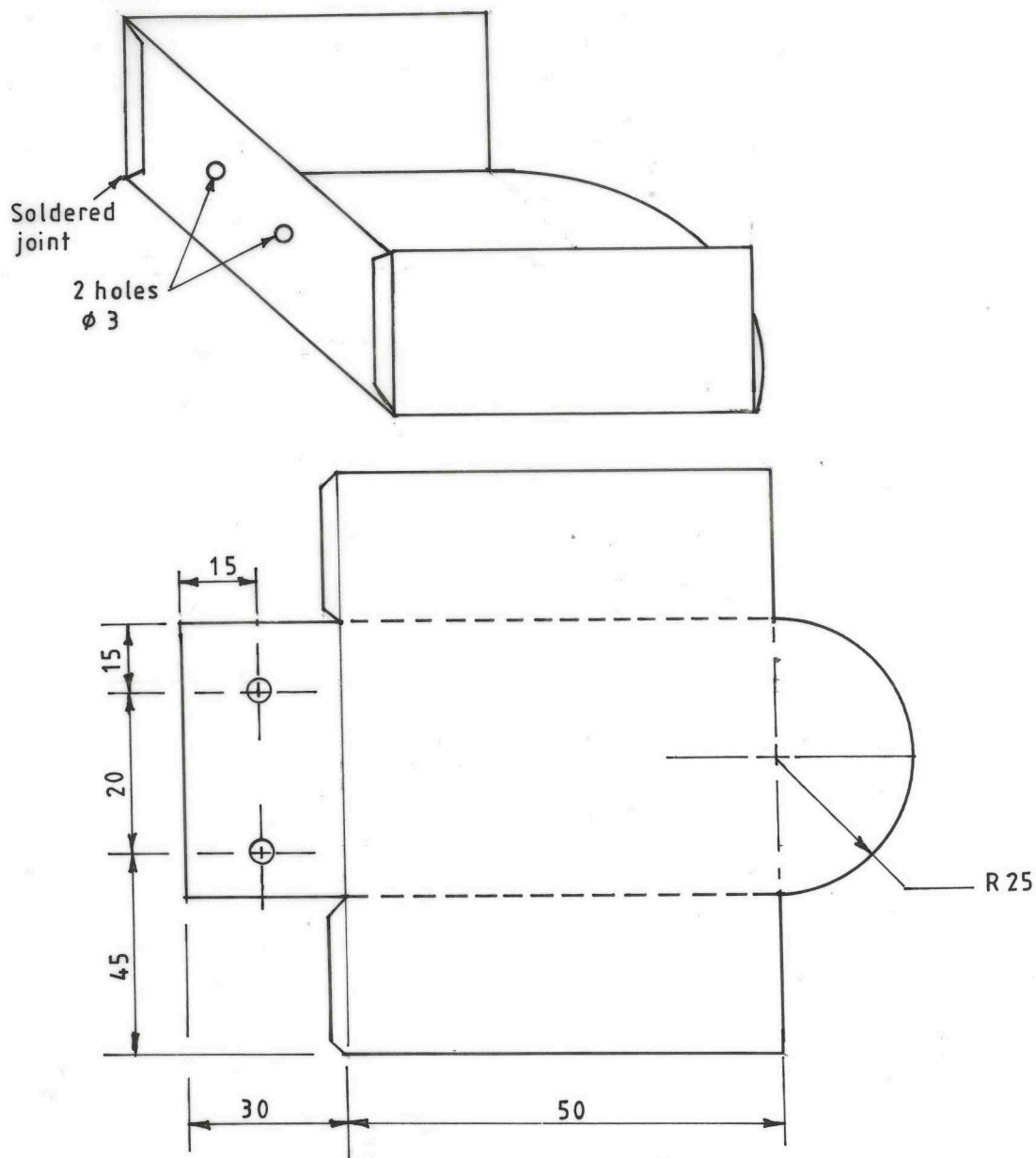
Power Mechanics Paper

STATION

- In the space below sketch in good proportion a sectional view of a mechanical fuel pump
Label four major parts

STATION

- Using the tools, equipment and materials provided make the scoop shown in figure 2



Figure



STATION

On the multi-plate clutch disc provided perform the following operations

1. measure and record the

depth of the four marked rivets marks

depth of the splines marks

3. Inspect the torsion spring for free play, damage and comment marks

4. Inspect the friction surface for its service condition and comment marks

Let the Examiner check your work

STATION

Identify the measuring tools labelled K to P and for each tool write the accuracy and two areas where it is used

TOOL	NAME	ACCURACY	USAGE
K			
L			
M			
N			
P			

STATION

Using the tools and materials provided connect a twin headlight parallel circuit controlled by a single switch marks

Let the Examiner check your work



STATION

Identify the tools and fasteners labelled A to J and state one use of each. (10 marks)

ITEM	NAME	USE
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		

STATION

On the single cylinder provided

(a) Demonstrate to the examiner how to check the roundness of the camshaft using a dial gauge

(b) Count the number of teeth on the crankshaft and the camshaft and calculate the gear ratio.

Crankshaft  teeth

Camshaft  teeth

Gear ratio  teeth

(10 marks)

STATION

Carry out a compression test on the single cylinder engine provided and record the reading in the space provided

Let the examiner check your work. (10 marks)

Compare your reading with the recommended reading provided by the examiner.

Comment on the state of the engine compression. (10 marks)

Comment



STATION

- Identify the parts labelled P to T on each part. Identify ONE defect and ONE possible effect on vehicle performance. Complete the table below.

PART	NAME	DEFECT	EFFECT
P			
Q			
R			
S			
T			

marks

STATION

- Using the tools and materials provided, perform the following operations on the mechanical fuel pump provided.

(a) Dismantle the fuel pump. (2 marks)

- Check the service condition of each of the following parts and comment on each.

Inlet valve marks

diaphragm marks

Assemble the pump and test it using the fuel provided. marks

Let the examiner check your work.